Expected Learning Outcomes

1. Understand how to deploy a React.js application to Azure Static Web Apps.
2. Learn how to deploy a .NET Web API to Azure App Service (Web App).
3. Integrate a front-end React app with a back-end API hosted on Azure.
4. Gain hands-on experience in managing and hosting full-stack applications in the cloud.

Task 1: Develop React and .NET Applications

1. Create two separate projects:
   - A **React.js front-end**.
   - A **.NET back-end Web API**.
2. Verify both applications run locally:
   - React app → http://localhost:3000/
   - .NET API → https://localhost:5001/ (or similar)
3. Push both projects to a GitHub repository:
4. git init
5. git add .
6. git commit -m "Initial commit"
7. git branch -M main
8. git remote add origin <your-github-repo-url>
9. git push -u origin main

Task 2: Configure Azure Resources

Step 1 – React Front-End (Azure Static Web Apps)

1. Log in to the Azure Portal.
2. Create a **Static Web App** resource.
   - Provide a **unique app name**.
   - Select the **GitHub repository** and branch containing the React code.
   - Choose **Free Plan** (if available).

- Configure build details:
  - **App location:** /
  - **Build output location:** build

3. Azure will **automatically create a GitHub Actions workflow (YAML)** in your repo under .github/workflows/.
   - This workflow builds and deploys your React app whenever you push changes.
4. Verify deployment: open the Azure Static Web App URL → your React app should load.

Step 2 – .NET Back-End (Azure App Service)

1. In Azure Portal, create a **Web App (App Service)**.
   - Provide a unique name.
   - Select **.NET 9 (or required version)** as the runtime stack.
   - Choose a **Free/Development plan**.
2. In the **Deployment Center**, connect the GitHub repository containing your .NET back-end code.
   - Azure will generate a **GitHub Actions workflow** to build and deploy your API automatically.
3. Verify deployment by visiting:

   https://<your-api-app-name>.azurewebsites.net/api/...

Submission : Submit a pdf file with the screenshots of each step of your work And the deployment link